Statistical Mechanics (MP461) Assignment 1 - Sample solutions

1. Quanta of energy

A system contains n atoms, each of which can only have zero or one quanta of energy. How many ways can you arrange r quanta of energy when (a) $n = 2, r = 1$; (b) $n = 20, r = 10$ and (c) $n = 2 \times 10^{23}$, $r = 1 \times 10^{23}$?

Solution

(a) $\binom{2}{1} = 2$

(b) $\binom{20}{10} = \frac{20}{10!10!} = 1.8 \times 10^5$

(c) Use Sterling $\ln n! \approx n \ln n - n$ to get

$$\begin{aligned} x = \ln \frac{2 \times 10^{23}!}{(10^{23})^2} &= 2 \times 10^{23} \ln(2 \times 10^{23}) - 2 \times 10^{23} - 2(10^{23} \ln 10^{23} - 10^{23}) \\ &= (2 \times 10^{23}) \ln \frac{2 \times 10^{23}}{1 \times 10^{23}} = (2 \times 10^{23}) \ln(2) \approx 1.38 \times 10^{23} \end{aligned}$$

so $x \approx e^{1.38 \times 10^2 3} = 10^y$. We have then that

$$\begin{aligned} \ln(x) = 1.38 \times 10^{23} &= \ln 10^y = y \ln 10 \\ \Rightarrow y &= \frac{\ln x}{\ln 10} \approx 1.38 \times 10^{23}/2.3 = 6 \times 10^{22} \end{aligned}$$

Final answer would then be

$$x = 10^{6 \times 10^{22}}$$

2. Fractional error

What fractional error do you make when using Stirling's approximation (in the form $\ln n! \approx n \ln n - n$) to evaluate (a) $\ln 10!$, (b) $\ln 100!$, and (c) $\ln 1000!$?

Solution The fractional error would be

$$F_E = \frac{\frac{1}{2} \ln 2\pi n}{n \ln n - n}$$

(a) $n = 10$ gives $F_E = 0.159$

(b) $n = 100$ gives $F_E = 0.0089$

(c) $n = 1000$ gives $F_E = 0.00074$

3. A fair die

(a) A throw of a regular die yields the numbers $1, 2, ..., 6$, each with probability $1/6$. Find the mean, variance, and standard deviation of the numbers obtained.

Solution

- Mean.
$$\langle x \rangle = \sum_i x_i P_i = (1+2+3+4+5+6)/6 = 3.5;$$

- Variance
$$\langle x^2 \rangle = \sum_i x_i^2 P_i = (1^2+2^2+3^2+4^2+5^2+6^2)/6 = 91/6;$$

  The variance is given is
$$Var(x) = \langle x^2 \rangle - \langle x \rangle^2 = 91/6 - 49/4 = 35/12.$$

  The standard deviation is the square root of this
$$\sigma(x) = \sqrt{Var(x)} \approx 1.71.$$

(b) Two throws of a die yield a total of 2 up to 12. What is the mean, variance, and standard deviation in this case?

Solution: Can work this out by brute force or realise that mean and variance are twice that of 1-throw and so s.d is $\sqrt{2}$ times.

- $\langle x \rangle = 7$
- Var$\approx 54.83 - 49 \approx 5.83$
- $s.d. \approx \sqrt{5.83} \approx 2.42$

(c) Calculate the probability of a total of 13 after 4 throws.

Solution One could enumerate all possibilities and count all the ways to get a 13. This is a bit tedious, however.

An easier way (I think) is to use the 2 throw probabilities. $P(2) = P(12) = 1/36$, $P(3) = P(11) = 2/36$ etc. .

So $P(13) = 2 \times (P(7)P(6) + ... + P(11)P(12)) \approx 0.1080$ where the factor of two accounts for the 2 ways each event could have occurred.

Is there another way? Up to permutations/order there are only a few (11) ways to make 13 with 4 dice. [6,5,1,1], [6,4,2,1], [6,3,3,1],[6,3,2,2], [5,5,2,1], [5,4,3,1],[5,4,2,2],[5,3,3,2], [4,4,3,2],[4,3,3,3],[4,4,4,1].

There are many ways it can happen for each of these combinations. We need to use the general combinatorial rule to work out how many ways for each case:

$$C = \frac{(N_{dice})!}{\prod_i (n_i!)}$$

where n_i is the number of times i occurs in a given unordered sequence. So for example for $[6, 5, 1, 1]$ we have

$$C = \frac{4!}{n_1!n_2!n_3!n_4!n_5!n_6!} = \frac{4!}{2!0!0!0!1!1!} = 12$$

Doing this for all 11 options above, one gets

Unordered sequence	combinations
6511	12
6421	24
6331	12
6322	12
5521	12
5431	24
5422	12
5332	12
4432	12
4333	4
4441	4

which amounts to 140 different ways to get 13. The probability is then $P(13) = 140/(6^4) \approx 0.1080$.

(d) What about the probability of a 14 after 4 throws?

Solution Same as before but we use
$P(14) = P(7)^2 + 2 \times (P(6)P(8) + ...P(2)P(12)) \approx 0.1127$

4. Poisson Distribution

This question is about a discrete probability distribution known as the Poisson distribution. Let x be a discrete random variable that can take the values $0, 1, 2, ...$ A quantity x is said to be Poisson distributed if the probability $P(x)$ of obtaining x is

$$P(x) = \frac{e^{-m}m^x}{x!} \tag{1}$$

where m is a particular number (which you will show in part (b) of this exercise is the mean value of x)

(a) Show that $P(x)$ is a well-behaved probability distribution in the sense that

$$\sum_{x=0}^{\infty} P(x) = 1 \tag{2}$$

(Why is this condition important?)

(b) Show that the mean value of the probability distribution is

$$\langle x \rangle = \sum_{x=0}^{\infty} xP(x) = m. \tag{3}$$

Solution

(a) We have

$$\sum_{x=0}^{\infty} \frac{e^{-m}m^x}{x!} = e^{-m}\sum_{x=0}^{\infty} \frac{m^x}{x!} = e^{-m}e^{m} = 1$$

Important because probabilities must sum to unity.

(b)

$$\langle x\rangle = \sum_{x=0} x\frac{e^{-m}m^x}{x!} = \sum_{x=1} x\frac{e^{-m}m^x}{x!} = \sum_{x=1} \frac{e^{-m}m^x}{(x-1)!} = m\sum_{x=1} \frac{e^{-m}m^{x-1}}{(x-1)!} = m\sum_{x=0} \frac{e^{-m}m^x}{(x)!} = m$$

where in the last equality we used part (a).

5. Exponential distribution

Let x be a continuous random variable that can take any value $x \geq 0$. A quantity is said to be exponentially distributed if it takes on values between x and $x+dx$ with probability:

$$P(x)dx = Ae^{-x/\lambda}dx, \tag{4}$$

where λ and A are constants.

(a) Find the value of A that makes $P(x)$ a well defined probability distribution.

(b) Show that the mean value of the probability distribution is λ.

(c) Find this probability distribution's variance and standard deviation.

Solution

(a)

$$\int P(x)dx = 1 \Rightarrow A\int_0^{\infty} e^{-x/\lambda}dx = -\lambda e^{-x/\lambda}\Big|_0^{\infty} = \lambda \Rightarrow A = 1/\lambda$$

(b)

$$\langle x\rangle = \frac{1}{\lambda}\int_0^{\infty} xP(x)dx = \frac{1}{\lambda}\int xe^{-x/\lambda}dx$$

We need to use integration by parts here $\int udv = uv - \int vdu$. Set $u = x$ and $dv = e^{-x/\lambda}dx$ so that $du = dx$ and $v = -\lambda e^{-x/\lambda}$ we then have

$$\frac{1}{\lambda}\int udv = \frac{1}{\lambda}\left(-\lambda xe^{-x/\lambda}\Big|_0^{\infty} + \lambda\int_0^{\infty} e^{-x/\lambda}dx\right) = \frac{1}{\lambda}\left(0 - \lambda^2 e^{-x/\lambda}\Big|_0^{\infty}\right) = \frac{1}{\lambda}\lambda^2 = \lambda$$

(c) We have $\langle x^2\rangle = \frac{1}{\lambda}\int_0^{\infty} x^2P(x)dx$. We need integration by parts again (twice!). Start with $u = x^2$ and $dv = e^{-x/\lambda}dx$ so that $du = dx$ and $v = -\lambda e^{-x/\lambda}$. We get

$$\begin{aligned}
\langle x^2\rangle = \frac{1}{\lambda}\int udv &= \frac{1}{\lambda}\left(-\lambda x^2 e^{-x/\lambda}\Big|_0^{\infty} + 2\lambda\int_0^{\infty} xe^{-x/\lambda}dx\right) \\
&= \left(-x^2 e^{-x/\lambda}\Big|_0^{\infty} + 2\int_0^{\infty} xe^{-x/\lambda}dx\right) \\
&= 2\lambda^2
\end{aligned}$$

where we used the calculation from part (b) in the last line. The standard deviation is therefore

$$\sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{2\lambda^2 - \lambda^2} = \lambda$$

6. Moment Generating Function

This question introduces an efficient method for calculating the mean and variance of probability distributions. We define the **moment generating function** $M(t)$ for a random variable x by

$$M(t) = \langle e^{tx} \rangle. \tag{5}$$

(a) Show that this definition implies that

$$\langle x^n \rangle = M^{(n)}(0), \tag{6}$$

where $M^{(n)}(t) = d^n M/dt^n$, the mean $\langle x \rangle = M^{(1)}(0)$ and the variance $\sigma_x = M^{(2)}(0) - [M^{(1)}(0)]^2$.

(b) Show that for the Poisson distribution, $M(t) = e^{m(e^t-1)}$ agrees with the mean calculated earlier.

(c) Use this method to work out the variance of the Poisson distribution.

Solution We have $M(t) = \langle e^{tx} \rangle = \sum_i e^{tx_i} P_i$.

(a) $M^n(t) = \frac{d^n M}{dt^n} = \sum_i x_i^n e^{tx_i} P_i \Rightarrow M^n(0) = \sum x_i^n P_i = \langle x^n \rangle$

(b) $M(t) = \sum_{x=0} e^{tx} \frac{e^{-m} m^x}{x!} = e^{-m} \sum_{x=0} \frac{(me^t)^x}{x!} = e^{-m} e^{me^t} = e^{m(e^t-1)}$.

(c) First moment:

$$M^1(0) = \left. \frac{dM(t)}{dt} \right|_0 = \left. me^{m(e^t-1)} \times e^t \right|_0 = m$$

Second moment

$$M^2(0) = \left. \frac{d\left(me^{m(e^t-1)} \times e^t\right)}{dt} \right|_0 = \left. me^{m(e^t-1)} e^t + m^2 e^t e^{m(e^t-1)} e^t \right|_0 = m + m^2$$

which means that $Var(x) = m + m^2 - m^2 = m$

7: Random Walk in One Dimension

A particle starts at position $x = 0$ and takes a random walk along the x-axis. At each step, the particle moves either $+1$ or -1 with equal probability (1/2 each).

(a) After 4 steps, what is the probability that the particle is at position $x = 0$?

(b) After 4 steps, what is the probability that the particle is at position $x = +2$?

(c) In general, after N steps, what is the probability $P(x, N)$ that the particle is at position x? (Assume N and x have the same parity, i.e., both even or both odd)

(d) Investigate the behavior of $P(x, N)$ for large N. What type of distribution does it approach, and what can you conclude about the typical distance travelled after many steps?

Hint: Use Stirling's approximation for large factorials. Also, for the expansion, you may find it useful that $\ln(1+\epsilon) \approx \epsilon - \frac{\epsilon^2}{2}$ *for small* ϵ.

solution

Part (a): Probability of being at $x = 0$ after 4 steps

To reach position $x = 0$ after 4 steps, we need an equal number of steps in each direction. If we take n_+ steps to the right and n_- steps to the left, then:

$$n_+ + n_- = 4 \tag{7}$$
$$n_+ - n_- = 0 \tag{8}$$

Solving these equations: $n_+ = n_- = 2$.
The number of ways to choose which 2 of the 4 steps go to the right is $\binom{4}{2}$.
Since each step has probability $1/2$ of going in either direction, the probability is:

$$P(0,4) = \binom{4}{2}\left(\frac{1}{2}\right)^4 = 6 \cdot \frac{1}{16} = \frac{3}{8} \tag{9}$$

Part (b): Probability of being at $x = +2$ after 4 steps

To reach position $x = +2$ after 4 steps:

$$n_+ + n_- = 4 \tag{10}$$
$$n_+ - n_- = 2 \tag{11}$$

Solving: $n_+ = 3$, $n_- = 1$.
The number of ways to choose which 3 of the 4 steps go to the right is $\binom{4}{3}$.
The probability is:

$$P(2,4) = \binom{4}{3}\left(\frac{1}{2}\right)^4 = 4 \cdot \frac{1}{16} = \frac{1}{4} \tag{12}$$

Part (d): Gaussian limit for large N

We want to understand the behavior of $P(x, N)$ for large N. Starting with:

$$P(x,N) = \frac{N!}{\left(\frac{N+x}{2}\right)!\left(\frac{N-x}{2}\right)!}\frac{1}{2^N} \tag{13}$$

Ok, lots of paths through this. They all start with taking the logarithm and using Stirling's approximation $\ln n! \approx n\ln n - n + \frac{1}{2}\ln(2\pi n)$:

Let $n_+ = \frac{N+x}{2}$ and $n_- = \frac{N-x}{2}$. Since $n_+ + n_- = N$, we get:

$$\ln P(x,N) = N\ln N - n_+\ln n_+ - n_-\ln n_- \tag{14}$$

$$+ \frac{1}{2}\ln(2\pi N) - \frac{1}{2}\ln(2\pi n_+) - \frac{1}{2}\ln(2\pi n_-) - N\ln 2 \tag{15}$$

For $x \ll N$, we expand $n_+ = \frac{N}{2}(1+\frac{x}{N})$ and $n_- = \frac{N}{2}(1-\frac{x}{N})$. Using $\ln(1+u) = u - \frac{u^2}{2} + O(u^3)$ and expanding systematically:

$$n_+\ln n_+ + n_-\ln n_- = N\ln\left(\frac{N}{2}\right) + \frac{x^2}{2N} + O\left(\frac{x^3}{N}\right) \tag{16}$$

Therefore:

$$N\ln N - n_+\ln n_+ - n_-\ln n_- = N\ln(2) - \frac{x^2}{2N} + O\left(\frac{x^3}{N}\right) \tag{17}$$

For the normalisation terms:

$$\frac{1}{2}\ln(2\pi N) - \frac{1}{2}\ln(2\pi n_+) - \frac{1}{2}\ln(2\pi n_-) \tag{18}$$

$$= \frac{1}{2}\ln(2\pi N) - \frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln(n_+ n_-) \tag{19}$$

$$= \frac{1}{2}\ln(2\pi N) - \ln(2\pi) - \frac{1}{2}\ln\left(\frac{N^2-x^2}{4}\right) \tag{20}$$

Expanding $\ln(N^2 - x^2) = 2\ln N - \frac{x^2}{N^2} + O(x^4/N^4)$ and simplifying:

$$= -\frac{1}{2}\ln(2\pi N) + \ln(2) + \frac{x^2}{2N^2} + O\left(\frac{x^4}{N^4}\right) \tag{21}$$

Combining everything and dropping higher-order terms:

$$\ln P(x,N) = N\ln(2) - \frac{x^2}{2N} - \frac{1}{2}\ln(2\pi N) + \ln(2) - N\ln(2) \tag{22}$$

$$= -\frac{x^2}{2N} - \frac{1}{2}\ln(2\pi N) + \ln(2) \tag{23}$$

Therefore:

$$P(x,N) \approx \sqrt{\frac{2}{\pi N}}\exp\left(-\frac{x^2}{2N}\right) \tag{24}$$

Key discovery: The distribution becomes Gaussian with width $\sqrt{N}$! The factor of $\sqrt{2}$ compared to the standard Gaussian normalisation arises because the discrete random walk can only reach positions with the same parity as N (i.e., we're effectively sampling every other integer).

This reveals the fundamental **diffusive scaling**: typical displacements grow as $\sqrt{N}$, not N. To go twice as far on average requires four times as many steps. This $\sqrt{N}$ scaling appears throughout statistical mechanics and explains why diffusion is much slower than ballistic motion.